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APPLICATION DEVELOPMENT (WBL)

LAB ASSIGNMENT 2

Industry Project Module Proposal

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Project Background

The growth of information technology has urged educational institutions to shift towards manual processes to more efficient and integrated web-based systems. Nevertheless, attendance management remains commonly addressed in fragmented or manual ways which may create inefficiencies, wrong records and challenges in tracking teaching activities and classes schedules. The absence of the combination of the attendance tracking and timetable management functions with reporting functions also makes the whole teaching and learning process more complex.

To solve these problems, a detailed web-based Attendance System is suggested to be the main part of the Teaching and Learning System. This system consolidates various features into one platform such as taking attendance, reporting discipline problems, creating attendance reports, uploading timetable data, seeing timetable schedules and scheduling replacement classes. All the components are intended to process a particular aspect of the attendance process, which creates an integrated and unified system of managing academic processes.

Moreover, the system will have the necessities of a base system like getting logged in, registering and managing users to have secure access and to be able to administer the users. The system is set up mainly to serve lecturers and administrators, with a capacity of 200 accounts of lecturers. With the implementation of this system, institutions are able to enhance efficiency, data accuracy, streamline operations and aid in the enhancement of monitoring and decision-making in teaching and learning management.

Target Users

The primary users of this application are Lecturers and Administrators, each with distinct roles:

- **Lecturer:** The main users of the application. They will use the app to view their daily timetables, mark student attendance for their classes, report any student discipline issues, and book or reschedule replacement classes.
- **Admin:** The administrative staff who manage the backend data. They are responsible for uploading the master time schedules into the system and viewing high-level statistical reports on overall attendance and student behavior.

Mandatory Modules

The system has three mandatory basic modules needed for it to function: Login, Registration and Manage User. A team member is responsible for the lead development of each module.

- Login - Enables users to log into the database. Led by *Chau Ying Jia*.
This module serves as the primary authentication gateway. It allows administrators to securely access the application using their registered credentials. It ensures that only authorized users can enter the database and interact with system features based on their specific roles.
- Registration - For new users (lecturer/admin) to sign up for an account with input validation. Led by *Gui Kah Sin*.
This module provides a structured sign-up process for new administrative users, incorporating input validation to ensure all account data is accurate and secure before being added to the system. It allows for the creation of new profiles while maintaining data integrity across the platform.
- Manage User - Allows the admin to manage user data, including updating of user details. Led by *Muhammad Naim bin Abdullah*.

Industry Modules

1. Taking Attendance (Lead: *Chau Ying Jia*)

The module aims to digitize and automate the tracking of student's attendance in various lectures. This module allows recording the attendance data which is then processed to generate reports that help administrators monitor student engagement throughout the semester.

2. Report Discipline Issue (Lead: *Gui Kah Sin*)

This module allows lecturers to record student discipline issues such as frequent absence or misconduct in a structured manner. It ensures that all cases are properly documented with relevant details for future reference. The system also supports categorization of issues and automatic detection of students with low attendance (below 80%), helping to highlight potential concerns early. Additionally, the collected data enables the administrators to monitor patterns, filter cases efficiently and take appropriate actions when necessary, improving overall student management.

3. Reporting Module (Lead: *Muhammad Naim Bin Abdullah*)

Provides visual statistics and data summaries regarding student attendance. It helps users quickly understand key metrics like overall attendance percentages and historical trends.

4. Upload Time Schedule (Lead: *Woo Cheng Shuan*)

The module will enable the administrators to input and update timetable information, such as the details of classes like subject, lecturer date and time. It also makes sure that the information of scheduling is saved in an organized way and that it can be easily updated in the event of any changes.

5. Showing Time Schedule (Lead: *Woo Cheng Shuan*)

This module presents the timetable of the lecturer in a structured format that enables the users to see the planned classes and mark attendance easily. It enhances the visibility of day to day teaching schedules and helps lecturers to manage their time better.

6. Booking Module (Lead: *Muhammad Mukhritz Al Iman bin Mohd Raffi*)

Provides a scheduling tool for lecturers to manage class adjustments and conflicts. It allows them to easily request to reschedule existing slots or book rooms for replacement classes.

Module Distribution Table

Module	Type	Lead Member
Login	Mandatory	Chau Ying Jia
Registration	Mandatory	Gui Kah Sin
Manage User	Mandatory	Muhammad Naim bin Abdullah
Taking Attendance	Industry	Chau Ying Jia
Report Discipline Issue	Industry	Gui Kah Sin
Reporting Module	Industry	Muhammad Naim bin Abdullah
Upload Time Schedule	Industry	Woo Cheng Shuan
Showing Time Schedule	Industry	Woo Cheng Shuan
Booking Module	Industry	Muhammad Mukhritz Al Iman bin Mohd Raffi

Justification

The module distribution was designed to ensure a fair workload while keeping functionally related features grouped together. To balance the foundational requirements with the core business logic, Ying Jia, Kah Sin , and Naim are each pairing one of the mandatory user management modules with the application's primary attendance and discipline tracking features. This setup allows them to integrate the necessary user access flows directly into the specific reporting systems they are responsible for building.

For the timetable operations, Cheng Shuan will manage both uploading the time schedules and displaying the timetable slots, ensuring the data structure remains consistent between the schedule's input and output views. Finally, Mukhritz is assigned exclusively to the Booking Module; because rescheduling and managing replacement classes requires complex conflict-checking and schedule manipulation, dedicating a single focus to this feature ensures the logic will be robust and well-executed.

Conclusion

In conclusion, the proposed web-based Attendance System provides a comprehensive solution to address the inefficiencies of conventional attendance and scheduling methods. By integrating key functionalities such as attendance tracking, discipline reporting, timetable management and reporting analytics into a single platform, the system reduces manual workload and improves overall operational efficiency.

Each module has been carefully designed and distributed among the team members to ensure balanced and smooth development as well as strong integration between features. The inclusion of both mandatory and industry modules ensures that the system meets basic functional requirements while also delivering practical value aligned with industry needs.

Ultimately, this system aims to support lecturers and administrators in managing students and organizing processes more effectively. With better data organization, improved monitoring capabilities and streamlined workflows, the system can lead to a more structured academic environment and informed decision-making.